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TECHNO-SOLUTIONISM

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THE BELIEF THAT "TECH" (ALONE) CAN SOLVE HARD SOCIAL PROBLEMS

ESPECIALLY IRRATIONAL EXUBERANCE THEREIN ...*

(*) <https://artearthtech.com/2017/09/18/techno-solutionism/>
<http://rufuspollock.com/2017/05/11/thoughts-on-blockchain-in-geneva/>
<https://medium.com/civic-tech-thoughts-from-joshdata/so-you-want-to-reform-democracy-7f3b1ef10597>

EXAMPLES (FOR DISCUSSION)

INTERNET WILL SOLVE EVERYTHING

INTERNET AND DEMOCRACY (EVERYONE KNOWS EVERYTHING NOW!); INTERNET AND MARKETS (NO FRICTION NOW!); INTERNET AND NEWS (ANYONE CAN BROADCAST NOW!)

BLOCKCHAIN WILL SOLVE EVERYTHING

FIXES CENTRALIZED CONTROL OF: MONEY, GOVERNMENT, BUSINESS, FACEBOOK, THE INTERNET. SOLVES FUNDAMENTAL PROBLEMS IN DOING DEMOCRATIC, DISTRIBUTED COORDINATION AT SCALE THAT WE'VE STRUGGLED WITH FOR 1000S OF YEARS

TECH WILL SOLVE CLIMATE CHANGE

IRON FILINGS IN THE OCEAN; HYDROGEN FUEL CELLS; FUSION, SOLAR ...

EXAMPLES (FOR DISCUSSION)

DOTCOM

BLOCKCHAIN & BITCOIN

ENVIRONMENTALISM

OZZIE ZEHNER (2013), GREEN ILLUSIONS: THE DIRTY SECRETS OF CLEAN ENERGY AND THE FUTURE OF ENVIRONMENTALISM

WALKS THROUGH EVERY MAJOR “CLEAN” ENERGY ALTERNATIVE FROM SOLAR TO NUCLEAR AND SHOWS FUNDAMENTAL LIMITATIONS AND/OR ENVIRONMENTAL COSTS ... => THERE IS NO SUSTAINABLE “CLEAN” WAY TO KEEP GOING THE WAY WE ARE

TECHNO-SOLUTIONISM

TECH WILL ALLOW US TO KEEP USING ENERGY IN SAME WAY AND BE CLEAN ... FALSE & ILLUSORY

AN ATTRACTIVE ILLUSION

TECHNO-SOLUTIONISM GIVES US AN EASY WAY OUT — NO CHANGE IS NEEDED

REAL ENVIRONMENTAL OPTIONS

1. WOMEN’S RIGHTS — REDUCE AND REVERSE POPULATION GROWTH THROUGH FEMALE EMPOWERMENT
2. IMPROVING CONSUMPTION — CONSUME LESS (NOT CONSUME “SUSTAINABLE”, AN OXYMORON)
3. THE ARCHITECTURE OF COMMUNITY — COMPACT, LIVABLE, WALKABLE CITIES
4. EFFICIENCY — DO MORE WITH LESS; OR THE SAME WITH MUCH LESS

NOTE

THESE ALL INVOLVE “INSTITUTIONAL / SOCIAL” INNOVATION (RATHER THAN TECH INNOVATION), DRIVEN BY AND SUPPORTING COLLECTIVE ACTION

COGNITIVE ORIGINS OF TECHNO-SOLUTIONISM

ANATOMY OF A SOLUTION

WE NEED A PROBLEM

IMAGINE IT IS SOLVED BY SOMETHING

A TECHNOLOGY THAT CAN DO ANYTHING ("MAGIC") COULD BE A SOLUTION

A SUFFICIENTLY ADVANCED TECH IS INDISTINGUISHABLE FROM MAGIC

CELL PHONES MIGHT AS WELL BE MAGIC TO MANY PEOPLE

SO ACTUALLY IS CURRENCY

MAYBE EVEN GROCERY STORES ...

CONTRIBUTING FACTORS

SURROUNDED BY MARVELS

MANY THINGS ARE EASY TO IMAGINE BUT HARD ACTUALLY TO ACHIEVE

GROCERY STORES IN THE DOT.COM ERA — A VERY ADVANCED TECH

IT IS ACTUALLY REALLY HARD TO REPLICATE THE GROCERY STORE EXPERIENCE ...

THE PROBLEM ZOO

*OR, WHEN IS A TECH A SOLUTION AND WHEN IS IT
TECHNO-SOLUTIONISM?*

GETTING TO THE ROOTS

WHAT ARE THE TYPES OF **PROBLEMS** WHERE TECH IS **NOT** THE MAIN ANSWER?

— WHAT ARE THE TYPES OF PROBLEM WHERE TECH **IS** THE (MAIN) ANSWER?

ARE THERE OTHER WAYS WE MISUNDERSTAND THE POTENTIAL IMPACT OF TECH?

4 TYPES OF PROBLEM

[HTTPS://ARTEARTHTECH.COM/2017/09/10/FOUR-TYPES-OF-PROBLEM/](https://artearthtech.com/2017/09/10/four-types-of-problem/)

There are 4 types of problem. Problems of:

1. **Science/tech**: knowing what the answer is (how does Global Warming happen, how do we prevent it?)
2. **Preference (or self-knowledge)**: knowing what we really want – do I want to watch this boxset or do meditation? Do I really want to eat this donut or do I really want this salad? Do I want to make sacrifices now to reduce carbon emissions so my daughter faces less risk from climate change in thirty years?
3. **Will (or self-control)**: we know what we want, but we struggle to act on it (I want to go on a diet but I don't; I want to avoid long-distance flights to save CO₂ but I really want to go on that holiday in the Caribbean)
4. **Collective (collective action)**: we need to act together to get it (I want to stop global warming by reducing carbon emissions but on my own I will have no impact – I need millions of others to act with me ...)

Associatedly we also have four types of solutions associated to each problem type – sciento-technic solutions, those targetting self-knowledge or will and finally those addressing collective action ("politico-economic").

4 TYPES OF PROBLEM

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All our techno-scientific advances in recent centuries have only helped with the first type of problem and we are currently suffering from a fetish of “techno-solutionism” where we mistakenly assume that all problems are of type one, or can be solved by type one solutions.

This is because type one problems are relatively easy: we can do science, build apps, etc. Solutions to the last three all involve deep questions of the human spirit and social organization. “Innovation” in these areas is very slow – for example, democracy (a solution on the 4th item) is basically unchanged since Ancient Athens.

Major progress on the last three must come from “**transformation of being**”. We do have answers here — they just aren’t easy to follow or “scale”. They are rooted in ontology and practices such as mindfulness found in traditions like Buddhism.

They require **will** and **culture** and are amenable to easy replication: just because you get enlightened does not mean I do — whereas, if you find a better way to grow corn it is easy for you to share that with me.

Unfortunately for us, science / tech solutions seem to help little with the other 3 items and as we get richer the value of science / tech solutions are less and less important ...

COLLECTIVE ACTION PROBLEM

PRINCIPAL AGENT PROBLEM

THESE ARE FUNDAMENTAL PROBLEMS FOR COORDINATION AND GROUP ACTION THAT
TECH MAKES LITTLE DIFFERENCE TO ...

GETTING TO THE ROOTS

WE UNDERESTIMATE OR IGNORE THE MEDIATING AND/OR CATALYTIC EFFECT OF SOCIALLY-DETERMINED RULES ON THE IMPACT OF TECH

TECH $\Rightarrow \Rightarrow \Rightarrow \Rightarrow \Rightarrow$ SOCIETY



RULES

EXAMPLE: THE OPEN REVOLUTION!